Huawei Access Node HSR-ACC-64 Failure Resolution Procedure for ALARM_6493: ASIC Memory Parity Errors, Line Card Crashes, and Power Supply Redundancy Loss

Node: HSR-ACC-64 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_6493

Description

The Access Node HSR-ACC-64, located at HSR Layout Access, is a Huawei-based Access device responsible for aggregating multiple Ethernet links and forwarding traffic to the core network. The ALARM_6493 indicates a hardware or environmental fault in the ASIC memory parity checking mechanism, line card failure, or loss of power supply redundancy. This failure domain occurs due to the intricate architecture of the Huawei Access Node's ASIC chip, which employs advanced memory parity checking to ensure data integrity. However, if the ASIC's memory parity checking mechanism fails or the line card crashes, it can lead to data corruption, packet loss, or even a complete network outage. Furthermore, the loss of power supply redundancy can also trigger the ALARM_6493, as the Access Node relies on dual power supplies to ensure continuous operation. Therefore, it is essential to investigate and resolve the root cause of the failure to prevent any potential data loss or network downtime.

Pre-requisite Checks

- Verify that the site is accessible and safe for remote hands to perform the necessary checks and repairs.

- Check the redundant link status to ensure that it is operational and not contributing to the failure.

- Gather system logs and alarm information to analyze the failure sequence and timing.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

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Command: display diagnostic information, Expected Output: Diagnostic information for the ASIC, including memory parity checking status and error counts., Description: This command provides a comprehensive view of the ASIC's diagnostic information, including memory parity checking status and error counts. It helps identify potential issues related to
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memory parity errors.

Command: display line-card information, Expected Output: Line card status, including operational state and current version., Description: This command provides detailed information about the line card's operational state and current version. It helps determine if the line card is functional and whether it may be contributing to the failure.

Command: display power supply information, Expected Output: Power supply status, including operational state and current capacity., Description: This command provides essential information about the power supply's operational state and current capacity. It helps determine if the power supply redundancy is intact and whether it may be contributing to the failure.
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Resolution Steps

Step 1: Step: Replace the Faulty ASIC, Command: replace asic [asic-id], Description: Replace the faulty ASIC with a new one to resolve the memory parity errors and prevent further data corruption.

Step 2: Step: Reboot the Line Card, Command: reboot line-card [line-card-id], Description: Reboot the line card to resolve the crash and restore normal operation. This step may require additional configuration to ensure seamless communication with the core network.

Step 3: Step: Verify Power Supply Redundancy, Command: verify power supply redundancy, Description: Verify that the power supply redundancy is intact and functioning correctly. If not, replace the faulty power supply unit to ensure continuous operation.

Post-Check Verification

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Command: display diagnostic information, Expected Output: Diagnostic information for the ASIC, including memory parity checking status and error counts., Description: Verify that the ASIC's diagnostic information shows no memory parity errors and that the error counts are zero.

Command: display line-card information, Expected Output: Line card status, including operational state and current version., Description: Verify that the line card is operational and functioning correctly.

Command: display power supply information, Expected Output: Power supply status, including operational state and current capacity., Description: Verify that the power supply redundancy is intact and functioning correctly.
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Roll-back Procedures

- Step: Restore Original Configuration, Command: restore original configuration, Description: Restore the original configuration to the device to ensure seamless communication with the core network. This step may require additional configuration to ensure smooth operation.

- Step: Revert Software Version, Command: revert software version, Description: Revert the software version to the previous version to ensure compatibility with the core network. This step may require additional configuration to ensure seamless communication.